

Final Business Analysis Churn

Hunter Davis

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Final Business Analysis Churn Data

```
# Load required libraries
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2    3.5.2      v tibble    3.2.1
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'caret'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## lift
```

```
library(rpart)
```

```
library(rpart.plot)
```

```
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
library(FactoMineR)
```

```
# 1. Data Loading and Exploration -----
```

```
data <- read.csv("~/AMU/Dats201/Week 8/Telecom_Customer_Churn_Dataset.csv")
```

```
str(data)
```

```
## 'data.frame': 1000 obs. of 10 variables:
## $ CustomerID : chr "ID0001" "ID0002" "ID0003" "ID0004" ...
## $ Age : int 56 69 46 32 60 25 38 56 36 40 ...
## $ AnnualIncome : num 56631 42124 43462 48864 31004 ...
## $ MonthlyCharges : num 65.9 43.7 119.3 113.4 78.6 ...
## $ TotalCharges : num 7745 6208 7631 5769 2735 ...
## $ TenureMonths : int 23 46 8 59 59 13 39 28 61 16 ...
## $ NumOfProducts : int 1 1 2 4 1 4 3 3 1 4 ...
## $ HasTechSupport : chr "No" "Yes" "No" "No" ...
## $ IsSeniorCitizen: int 0 1 0 0 0 1 0 0 1 1 ...
## $ Churn : chr "Yes" "Yes" "Yes" "Yes" ...
```

```
summary(data)
```

```
## CustomerID Age AnnualIncome MonthlyCharges
## Length:1000 Min. :18.00 Min. :30232 Min. : 30.01
## Class :character 1st Qu.:31.00 1st Qu.:41857 1st Qu.: 53.34
## Mode :character Median :44.00 Median :55215 Median : 75.38
## Mean :43.82 Mean :54910 Mean : 74.95
## 3rd Qu.:56.00 3rd Qu.:66907 3rd Qu.: 97.67
## Max. :69.00 Max. :79971 Max. :119.85
## TotalCharges TenureMonths NumOfProducts HasTechSupport
## Min. : 500.1 Min. : 1.00 Min. :1.000 Length:1000
## 1st Qu.:2362.9 1st Qu.:18.75 1st Qu.:2.000 Class :character
## Median :4252.9 Median :36.00 Median :2.000 Mode :character
## Mean :4239.0 Mean :35.89 Mean :2.467
## 3rd Qu.:6115.9 3rd Qu.:53.00 3rd Qu.:3.000
## Max. :7991.7 Max. :71.00 Max. :4.000
## IsSeniorCitizen Churn
## Min. :0.000 Length:1000
## 1st Qu.:0.000 Class :character
## Median :0.000 Mode :character
## Mean :0.497
## 3rd Qu.:1.000
## Max. :1.000
```

```
head(data)
```

```
## CustomerID Age AnnualIncome MonthlyCharges TotalCharges TenureMonths
## 1 ID0001 56 56630.86 65.94 7745.36 23
## 2 ID0002 69 42123.53 43.72 6207.78 46
## 3 ID0003 46 43462.16 119.32 7630.80 8
## 4 ID0004 32 48864.21 113.43 5769.05 59
## 5 ID0005 60 31003.56 78.60 2735.39 59
## 6 ID0006 25 46103.96 105.78 1290.24 13
## NumOfProducts HasTechSupport IsSeniorCitizen Churn
## 1 1 No 0 Yes
## 2 1 Yes 1 Yes
## 3 2 No 0 Yes
## 4 4 No 0 Yes
## 5 1 Yes 0 Yes
## 6 4 No 1 Yes
```

```
colnames(data)
```

```
## [1] "CustomerID"      "Age"              "AnnualIncome"     "MonthlyCharges"
## [5] "TotalCharges"    "TenureMonths"     "NumOfProducts"    "HasTechSupport"
## [9] "IsSeniorCitizen" "Churn"
```

```
colSums(is.na(data))
```

```
##      CustomerID      Age      AnnualIncome      MonthlyCharges      TotalCharges
##           0           0           0           0           0
##      TenureMonths      NumOfProducts      HasTechSupport      IsSeniorCitizen      Churn
##           0           0           0           0           0
```

```
data$CustomerID <- NULL
```

```
# Convert categorical variables to factors
```

```
data$HasTechSupport <- as.factor(data$HasTechSupport)
```

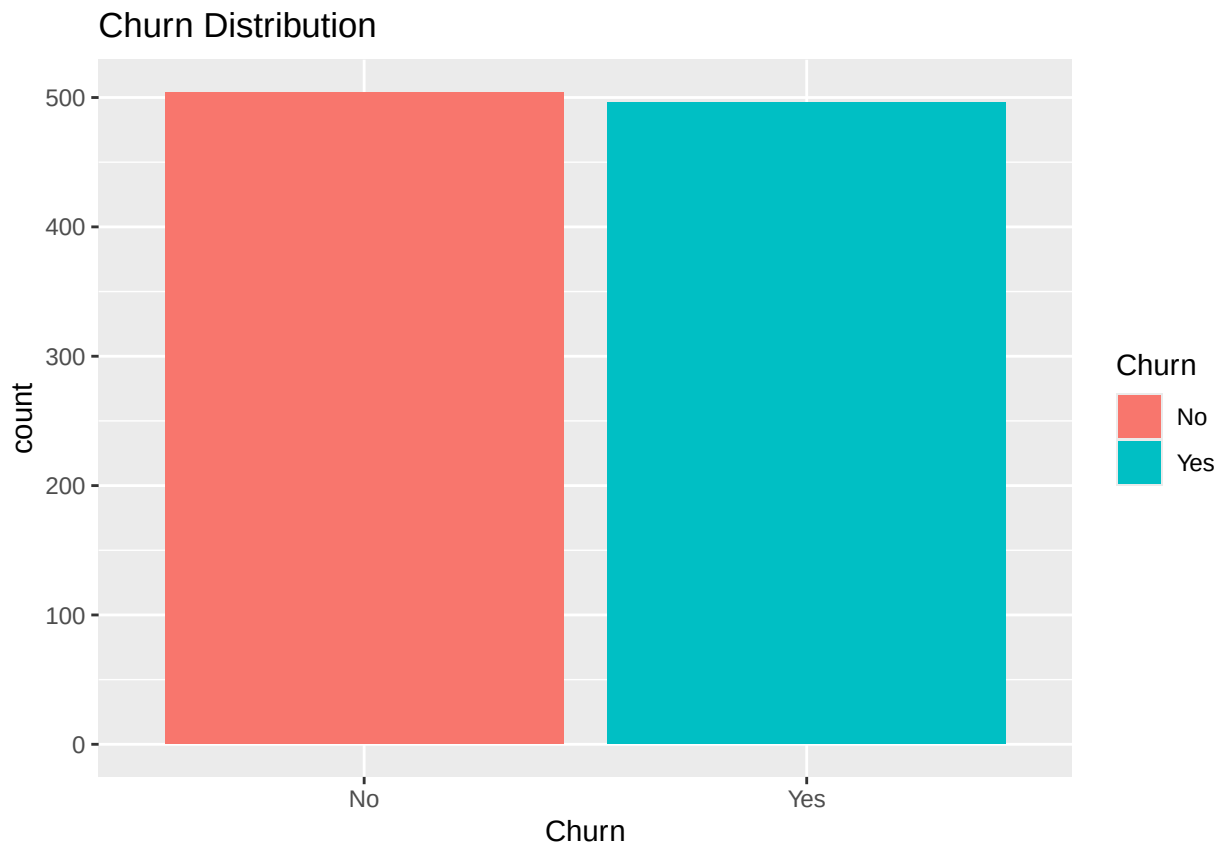
```
data$IsSeniorCitizen <- as.factor(data$IsSeniorCitizen)
```

```
data$Churn <- as.factor(data$Churn)
```

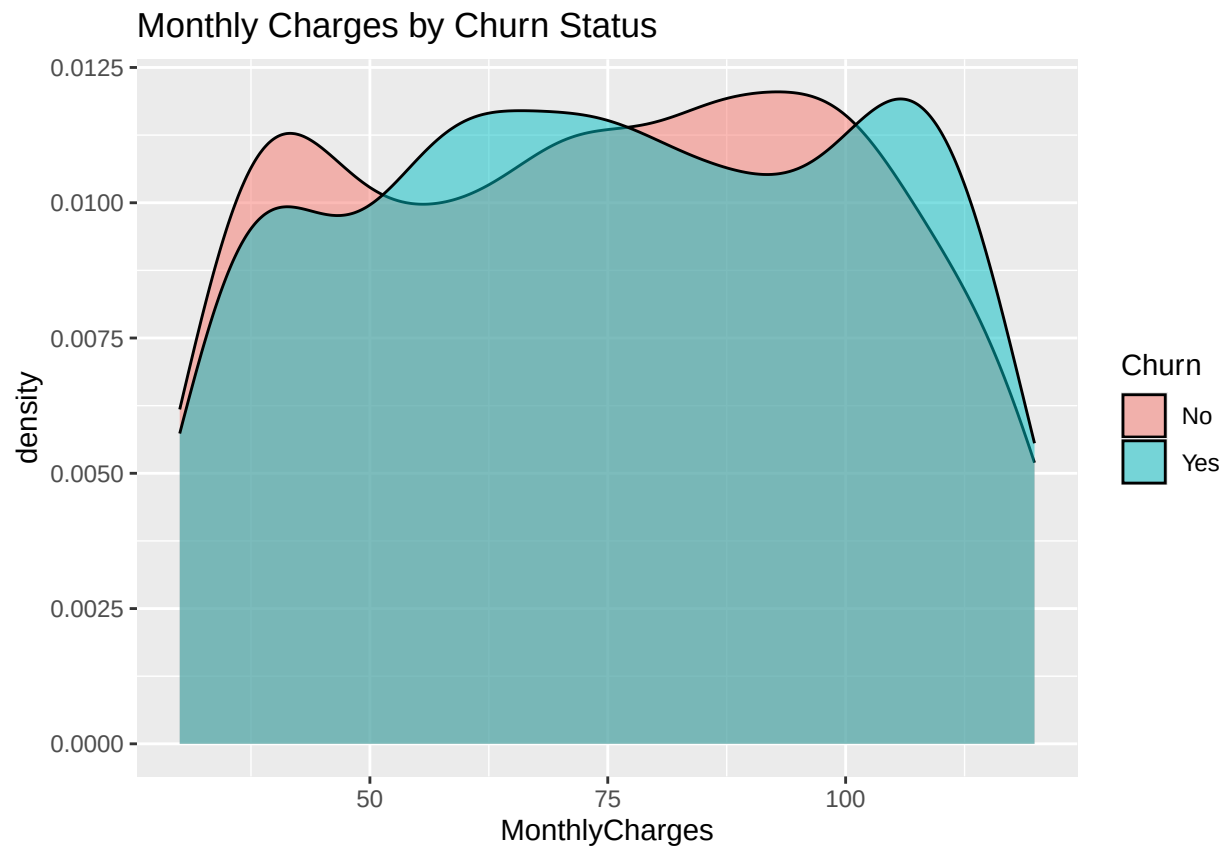
```
data$NumOfProducts <- as.factor(data$NumOfProducts)
```

```
# Data Visualization
```

```
ggplot(data, aes(x=Churn, fill=Churn)) +  
  geom_bar() +  
  ggtitle("Churn Distribution")
```



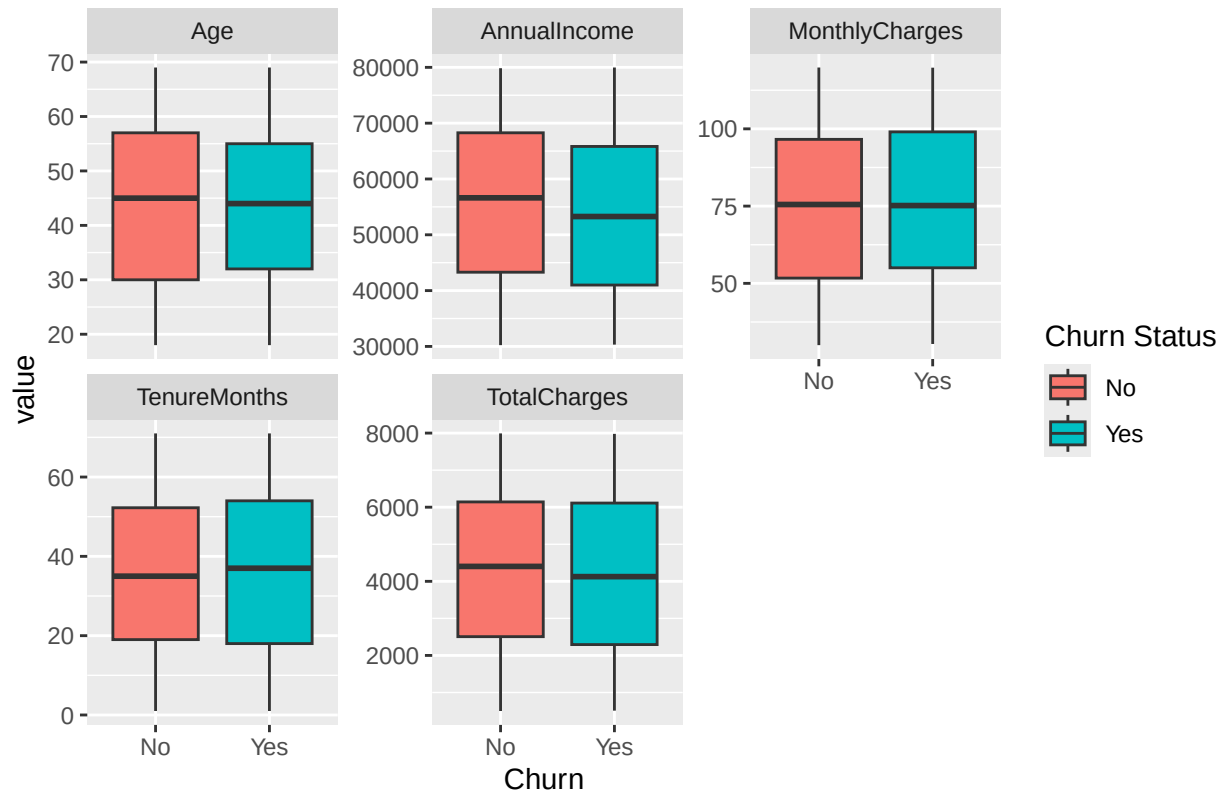
```
ggplot(data, aes(x=MonthlyCharges, fill=Churn)) +
  geom_density(alpha=0.5) +
  ggtitle("Monthly Charges by Churn Status")
```



```
numeric_vars <- c("Age",
                  "AnnualIncome",
                  "MonthlyCharges",
                  "TotalCharges",
                  "TenureMonths")

data %>%
  pivot_longer(cols = all_of(numeric_vars), names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = Churn, y = value, fill = Churn)) +
  geom_boxplot() +
  facet_wrap(~ variable, scales = "free_y") +
  labs(title = "Numeric Variables by Churn Status",
       x = "Churn",
       fill = "Churn Status")
```

Numeric Variables by Churn Status



Key Findings (EDA):

EDA

- No missing values ($\text{colSums}(\text{is.na}(\text{data})) = 0$ for all columns)
- Balanced Churn rate: 51% Yes, 49% No
- Numeric variable ranges:
 - Age: 18-69 years (mean 43.82)
 - Annual Income: \$30232-\$79971 (mean \$54910)
 - Monthly Charges: \$30.01-\$119.85 (mean \$74.95)
 - Total Charges: \$500-\$7991.70 (mean \$4239)
 - Tenure: 1-71 Months (mean 35.89)

Preprocessing

- Converted categorical variables to factors:
 - HasTechSupport (No/Yes)
 - IsSeniorCitizen (0/1)
 - Churn (No/Yes)
 - NumOfProducts (1-4)

```
# 2. Predictive Modeling -----
```

```
# Split data
```

```
set.seed(1)
```

```
trainIndex <- createDataPartition(data$Churn, p=0.7, list=FALSE)
```

```
train <- data[trainIndex,]
```

```
test <- data[-trainIndex,]
```

```
# Logistic Regression
```

```
logit_model <- glm(Churn ~ ., data=train, family="binomial")
```

```
summary(logit_model)
```

```
##
```

```
## Call:
```

```
## glm(formula = Churn ~ ., family = "binomial", data = train)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    7.159e-01  5.070e-01   1.412   0.1580
## Age            -2.397e-03  5.132e-03  -0.467   0.6405
## AnnualIncome   -1.366e-05  5.322e-06  -2.567   0.0103 *
## MonthlyCharges  1.689e-03  2.907e-03   0.581   0.5611
## TotalCharges   -3.102e-05  3.506e-05  -0.885   0.3764
## TenureMonths    1.287e-03  3.837e-03   0.335   0.7373
## NumOfProducts2 -6.651e-02  2.111e-01  -0.315   0.7527
## NumOfProducts3 -7.139e-02  2.149e-01  -0.332   0.7398
## NumOfProducts4 -2.064e-01  2.213e-01  -0.932   0.3511
## HasTechSupportYes 1.749e-02  1.536e-01   0.114   0.9094
## IsSeniorCitizen1 3.274e-01  1.532e-01   2.137   0.0326 *
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
```

```
## Null deviance: 971.76 on 700 degrees of freedom
```

```
## Residual deviance: 958.38 on 690 degrees of freedom
```

```
## AIC: 980.38
```

```
##
```

```
## Number of Fisher Scoring iterations: 4
```

```
# Decision Tree
```

```
tree_model <- rpart(Churn ~ ., data=train, method="class")
```

```
summary(tree_model)
```

```
## Call:
```

```
## rpart(formula = Churn ~ ., data = train, method = "class")
```

```
## n= 701
```

```
##
```

```
##           CP nsplit rel error    xerror      xstd
## 1 0.11206897    0 1.0000000 1.0890805 0.03791485
## 2 0.03017241    1 0.8879310 0.9626437 0.03800360
## 3 0.02298851    3 0.8275862 0.9540230 0.03798779
## 4 0.02011494    4 0.8045977 0.9367816 0.03794776
```

```

## 5 0.01149425      5 0.7844828 0.9568966 0.03799337
## 6 0.01005747      9 0.7327586 0.9712644 0.03801661
## 7 0.01000000     11 0.7126437 0.9683908 0.03801258
##
## Variable importance
##   TotalCharges   AnnualIncome           Age MonthlyCharges   TenureMonths
##             36             32             19             10             2
##
## Node number 1: 701 observations,      complexity param=0.112069
##   predicted class=No   expected loss=0.4964337   P(node) =1
##   class counts:   353   348
##   probabilities: 0.504 0.496
##   left son=2 (448 obs) right son=3 (253 obs)
##   Primary splits:
##     AnnualIncome < 48664.03 to the right, improve=5.148811, (0 missing)
##     MonthlyCharges < 104.345 to the left, improve=3.033518, (0 missing)
##     TotalCharges < 868.12 to the right, improve=2.785289, (0 missing)
##     IsSeniorCitizen splits as LR, improve=2.314623, (0 missing)
##     TenureMonths < 65.5 to the right, improve=1.904556, (0 missing)
##   Surrogate splits:
##     TotalCharges < 512.995 to the right, agree=0.645, adj=0.016, (0 split)
##
## Node number 2: 448 observations,      complexity param=0.03017241
##   predicted class=No   expected loss=0.4508929   P(node) =0.639087
##   class counts:   246   202
##   probabilities: 0.549 0.451
##   left son=4 (424 obs) right son=5 (24 obs)
##   Primary splits:
##     TotalCharges < 931.515 to the right, improve=3.361298, (0 missing)
##     MonthlyCharges < 104.025 to the left, improve=2.457468, (0 missing)
##     Age < 48.5 to the right, improve=1.748942, (0 missing)
##     AnnualIncome < 56272.85 to the left, improve=1.707951, (0 missing)
##     TenureMonths < 1.5 to the left, improve=1.349490, (0 missing)
##
## Node number 3: 253 observations,      complexity param=0.02011494
##   predicted class=Yes expected loss=0.4229249   P(node) =0.360913
##   class counts:   107   146
##   probabilities: 0.423 0.577
##   left son=6 (25 obs) right son=7 (228 obs)
##   Primary splits:
##     Age < 24.5 to the left, improve=2.614422, (0 missing)
##     AnnualIncome < 46512.48 to the left, improve=2.435074, (0 missing)
##     TotalCharges < 2954.725 to the right, improve=1.923592, (0 missing)
##     MonthlyCharges < 99.3 to the left, improve=1.658137, (0 missing)
##     IsSeniorCitizen splits as LR, improve=1.490696, (0 missing)
##   Surrogate splits:
##     TotalCharges < 507.525 to the left, agree=0.909, adj=0.08, (0 split)
##
## Node number 4: 424 observations,      complexity param=0.03017241
##   predicted class=No   expected loss=0.4363208   P(node) =0.6048502
##   class counts:   239   185
##   probabilities: 0.564 0.436
##   left son=8 (339 obs) right son=9 (85 obs)
##   Primary splits:

```

```

##      MonthlyCharges < 104.025  to the left,  improve=3.504649, (0 missing)
##      Age             < 34.5    to the right, improve=2.096733, (0 missing)
##      TotalCharges    < 2251.22 to the left,  improve=1.925856, (0 missing)
##      AnnualIncome     < 56272.85 to the left, improve=1.744886, (0 missing)
##      TenureMonths     < 59.5    to the right, improve=1.216796, (0 missing)
##      Surrogate splits:
##      AnnualIncome < 79849.97 to the left,  agree=0.802, adj=0.012, (0 split)
##
## Node number 5: 24 observations
##   predicted class=Yes  expected loss=0.2916667  P(node) =0.0342368
##   class counts:      7    17
##   probabilities: 0.292 0.708
##
## Node number 6: 25 observations
##   predicted class=No   expected loss=0.36  P(node) =0.03566334
##   class counts:      16    9
##   probabilities: 0.640 0.360
##
## Node number 7: 228 observations,    complexity param=0.01149425
##   predicted class=Yes  expected loss=0.3991228  P(node) =0.3252496
##   class counts:      91   137
##   probabilities: 0.399 0.601
##   left son=14 (207 obs) right son=15 (21 obs)
##   Primary splits:
##   AnnualIncome < 46512.48 to the left,  improve=3.038048, (0 missing)
##   IsSeniorCitizen splits as LR,          improve=1.979802, (0 missing)
##   TotalCharges < 2954.725 to the right, improve=1.605681, (0 missing)
##   Age < 53.5 to the right, improve=1.462133, (0 missing)
##   MonthlyCharges < 99.3 to the left, improve=1.035234, (0 missing)
##
## Node number 8: 339 observations
##   predicted class=No   expected loss=0.4041298  P(node) =0.4835949
##   class counts:      202   137
##   probabilities: 0.596 0.404
##
## Node number 9: 85 observations,    complexity param=0.02298851
##   predicted class=Yes  expected loss=0.4352941  P(node) =0.1212553
##   class counts:      37    48
##   probabilities: 0.435 0.565
##   left son=18 (16 obs) right son=19 (69 obs)
##   Primary splits:
##   TotalCharges < 2571.125 to the left,  improve=3.904177, (0 missing)
##   AnnualIncome < 76653.37 to the left,  improve=3.346677, (0 missing)
##   MonthlyCharges < 119.4 to the right, improve=2.714975, (0 missing)
##   Age < 34.5 to the right, improve=1.374442, (0 missing)
##   TenureMonths < 11.5 to the left, improve=1.187503, (0 missing)
##
## Node number 14: 207 observations,    complexity param=0.01149425
##   predicted class=Yes  expected loss=0.4251208  P(node) =0.2952924
##   class counts:      88   119
##   probabilities: 0.425 0.575
##   left son=28 (18 obs) right son=29 (189 obs)
##   Primary splits:
##   AnnualIncome < 44303.34 to the right, improve=3.480331, (0 missing)

```

```

##      TotalCharges    < 2954.725 to the right, improve=1.860225, (0 missing)
##      Age              < 53.5      to the right, improve=1.681768, (0 missing)
##      IsSeniorCitizen splits as LR,      improve=1.587015, (0 missing)
##      MonthlyCharges  < 99.3      to the left,  improve=1.075511, (0 missing)
##
## Node number 15: 21 observations
##   predicted class=Yes expected loss=0.1428571 P(node) =0.0299572
##   class counts:      3      18
##   probabilities: 0.143 0.857
##
## Node number 18: 16 observations
##   predicted class=No  expected loss=0.25 P(node) =0.02282454
##   class counts:      12      4
##   probabilities: 0.750 0.250
##
## Node number 19: 69 observations
##   predicted class=Yes expected loss=0.3623188 P(node) =0.09843081
##   class counts:      25      44
##   probabilities: 0.362 0.638
##
## Node number 28: 18 observations
##   predicted class=No  expected loss=0.2777778 P(node) =0.0256776
##   class counts:      13      5
##   probabilities: 0.722 0.278
##
## Node number 29: 189 observations, complexity param=0.01149425
##   predicted class=Yes expected loss=0.3968254 P(node) =0.2696148
##   class counts:      75     114
##   probabilities: 0.397 0.603
##   left son=58 (68 obs) right son=59 (121 obs)
##   Primary splits:
##     Age              < 53.5      to the right, improve=2.2613140, (0 missing)
##     AnnualIncome     < 31762.39 to the right, improve=2.0119050, (0 missing)
##     TotalCharges     < 2954.725 to the right, improve=0.9731320, (0 missing)
##     TenureMonths     < 14.5      to the left,  improve=0.8826421, (0 missing)
##     MonthlyCharges   < 62.13    to the right, improve=0.7505460, (0 missing)
##   Surrogate splits:
##     AnnualIncome < 30519.16 to the left,  agree=0.661, adj=0.059, (0 split)
##     TotalCharges < 7508.715 to the right, agree=0.661, adj=0.059, (0 split)
##     TenureMonths < 60.5      to the right, agree=0.651, adj=0.029, (0 split)
##
## Node number 58: 68 observations, complexity param=0.01149425
##   predicted class=No  expected loss=0.5 P(node) =0.09700428
##   class counts:      34      34
##   probabilities: 0.500 0.500
##   left son=116 (48 obs) right son=117 (20 obs)
##   Primary splits:
##     TotalCharges     < 6123.63 to the left,  improve=3.5416670, (0 missing)
##     Age              < 59.5      to the left,  improve=3.2196970, (0 missing)
##     TenureMonths     < 40.5      to the left,  improve=2.4010460, (0 missing)
##     MonthlyCharges   < 41.92    to the left,  improve=1.6007530, (0 missing)
##     AnnualIncome     < 41442.66 to the right, improve=0.9129968, (0 missing)
##   Surrogate splits:
##     Age              < 68.5      to the left,  agree=0.750, adj=0.15, (0 split)

```

```

##      AnnualIncome < 30571.63 to the right, agree=0.735, adj=0.10, (0 split)
##
## Node number 59: 121 observations
##   predicted class=Yes   expected loss=0.338843   P(node) =0.1726106
##   class counts:      41      80
##   probabilities: 0.339 0.661
##
## Node number 116: 48 observations,      complexity param=0.01005747
##   predicted class=No   expected loss=0.3958333   P(node) =0.06847361
##   class counts:      29      19
##   probabilities: 0.604 0.396
##   left son=232 (10 obs) right son=233 (38 obs)
##   Primary splits:
##     Age < 56.5 to the left, improve=2.2109650, (0 missing)
##     AnnualIncome < 34806.41 to the right, improve=1.8189390, (0 missing)
##     TotalCharges < 2944.405 to the right, improve=1.8184730, (0 missing)
##     TenureMonths < 40.5 to the left, improve=0.7440476, (0 missing)
##     MonthlyCharges < 86.965 to the right, improve=0.6297619, (0 missing)
##   Surrogate splits:
##     TotalCharges < 963.795 to the left, agree=0.833, adj=0.2, (0 split)
##     AnnualIncome < 30954.54 to the left, agree=0.812, adj=0.1, (0 split)
##
## Node number 117: 20 observations
##   predicted class=Yes   expected loss=0.25   P(node) =0.02853067
##   class counts:      5      15
##   probabilities: 0.250 0.750
##
## Node number 232: 10 observations
##   predicted class=No   expected loss=0.1   P(node) =0.01426534
##   class counts:      9      1
##   probabilities: 0.900 0.100
##
## Node number 233: 38 observations,      complexity param=0.01005747
##   predicted class=No   expected loss=0.4736842   P(node) =0.05420827
##   class counts:      20      18
##   probabilities: 0.526 0.474
##   left son=466 (21 obs) right son=467 (17 obs)
##   Primary splits:
##     TotalCharges < 2944.405 to the right, improve=3.3171160, (0 missing)
##     MonthlyCharges < 44.915 to the left, improve=1.8782440, (0 missing)
##     AnnualIncome < 42326.52 to the right, improve=1.8782440, (0 missing)
##     NumOfProducts splits as LLRL, improve=0.8784029, (0 missing)
##     TenureMonths < 40.5 to the left, improve=0.4360048, (0 missing)
##   Surrogate splits:
##     MonthlyCharges < 84.22 to the left, agree=0.658, adj=0.235, (0 split)
##     AnnualIncome < 33319.62 to the right, agree=0.632, adj=0.176, (0 split)
##     TenureMonths < 33.5 to the left, agree=0.632, adj=0.176, (0 split)
##     Age < 59.5 to the left, agree=0.605, adj=0.118, (0 split)
##     NumOfProducts splits as LLLR, agree=0.579, adj=0.059, (0 split)
##
## Node number 466: 21 observations
##   predicted class=No   expected loss=0.2857143   P(node) =0.0299572
##   class counts:      15      6
##   probabilities: 0.714 0.286

```

```
rpart.plot(tree_model, main="Decision Tree for Customer Churn")
```

```

graph TD
    Root["No  
0.50  
100%"] -- yes --> Node1["No  
0.45  
64%"]
    Root -- no --> Node2["Yes  
0.58  
36%"]
    
    Node2 --> Node3["Age < 25"]
    Node3 -- yes --> Node4["Yes  
0.60  
33%"]
    Node3 -- no --> Node5["AnnualIncome < 47e+3"]
    Node5 -- yes --> Node6["Yes  
0.57  
30%"]
    Node5 -- no --> Node7["AnnualIncome >= 44e+3"]
    Node7 -- yes --> Node8["Yes  
0.60  
27%"]
    Node7 -- no --> Node9["Age >= 54"]
    Node9 -- yes --> Node10["Yes  
0.60  
27%"]
    Node9 -- no --> Node11["TotalCharges < 6124"]
    Node11 -- yes --> Node12["No  
0.40  
7%"]
    Node11 -- no --> Node13["Age < 57"]
    Node13 -- yes --> Node14["No  
0.40  
7%"]
    Node13 -- no --> Node15["TotalCharges >= 2944"]
    Node15 -- yes --> Node16["Yes  
0.71  
2%"]
    Node15 -- no --> Node17["No  
0.29  
3%"]
    Node12 --> Leaf1["No  
0.40  
48%"]
    Node14 --> Leaf2["No  
0.25  
2%"]
    Node16 --> Leaf3["Yes  
0.64  
10%"]
    Node17 --> Leaf4["Yes  
0.71  
3%"]
    Node18["TotalCharges < 2571"] --> Leaf5["Yes  
0.36  
4%"]
    Node19["TotalCharges >= 932"] --> Leaf6["No  
0.28  
3%"]
    Node20["MonthlyCharges < 104"] --> Leaf7["No  
0.10  
1%"]
    Node21["MonthlyCharges >= 104"] --> Leaf8["No  
0.29  
3%"]
    Node22["TotalCharges < 2571"] --> Leaf9["Yes  
0.75  
3%"]
    Node23["TotalCharges >= 932"] --> Leaf10["Yes  
0.66  
17%"]
    Node24["AnnualIncome < 47e+3"] --> Leaf11["Yes  
0.86  
3%"]
  
```

Logistic Regression

- 11

- Estimate = 0.3274 (log-odds)
- Odds Ratio = $\exp(0.3274) = 1.387$
- Interpretation: Seniors have 38.7% higher churn odds than non-seniors
- Non-Significant Predictors ($p < 0.05$)
 - Age ($p=0.6405$): No significant age effect after accounting for other factors
 - Monthly Charges ($p=0.5611$): Pricing not significant when income is considered
 - Tenure Months ($p=0.7373$): Surprisingly, tenure length doesn't independently predict churn
 - Number of Products: All categories (2,3,4) vs baseline (1) show no significant differences ($p=0.75, 0.74, 0.35$)
 - Tech Support ($p=0.9094$): No protective effect found The logistic Model shows moderate performance with an AIC of 980.38 and residual deviance of 958.38 on 690 degrees of freedom.

Decision Tree

- Key Splits
 - Primary Split:
 - * Annual Income < \$48,664 (Right branch: 253 obs, 57.7% churn)
 - * Higher-income customers (>\$48K) have lower churn risk (45.1% churn)
 - Secondary Split:
 - * Age < 24.5: Younger customers in high-income group show 64% retention
 - * Total Charges < \$931: Customers with lower cumulative spending churn more
 - * Monthly Charges < \$104: Pricing sensitivity appears at this threshold
 - Deep Splits:
 - * Node 15: Customers with Age ≥ 53.5 & Income < \$46.5K have 85.7% churn rate
 - * Node 232: Customers with Age < 56.5 & Total Charges < \$2,944 show 90% retention
- Variable importance:
 - Total Charges (36%)
 - Annual Income (32%)
 - Age (19%)
 - Monthly Charges (10%)
 - Tenure (2%)

```
# Resampling Methods -----

# Set up cross-validation
ctrl <- trainControl(method = "cv",
                     number = 10,
                     savePredictions = TRUE,
                     classProbs = TRUE,
                     summaryFunction = twoClassSummary)

# Logistic Regression with CV
logit_cv <- train(Churn ~ ., data=train,
                 method = "glm",
                 family = "binomial",
                 trControl = ctrl,
                 metric = "ROC")

print(logit_cv)
```

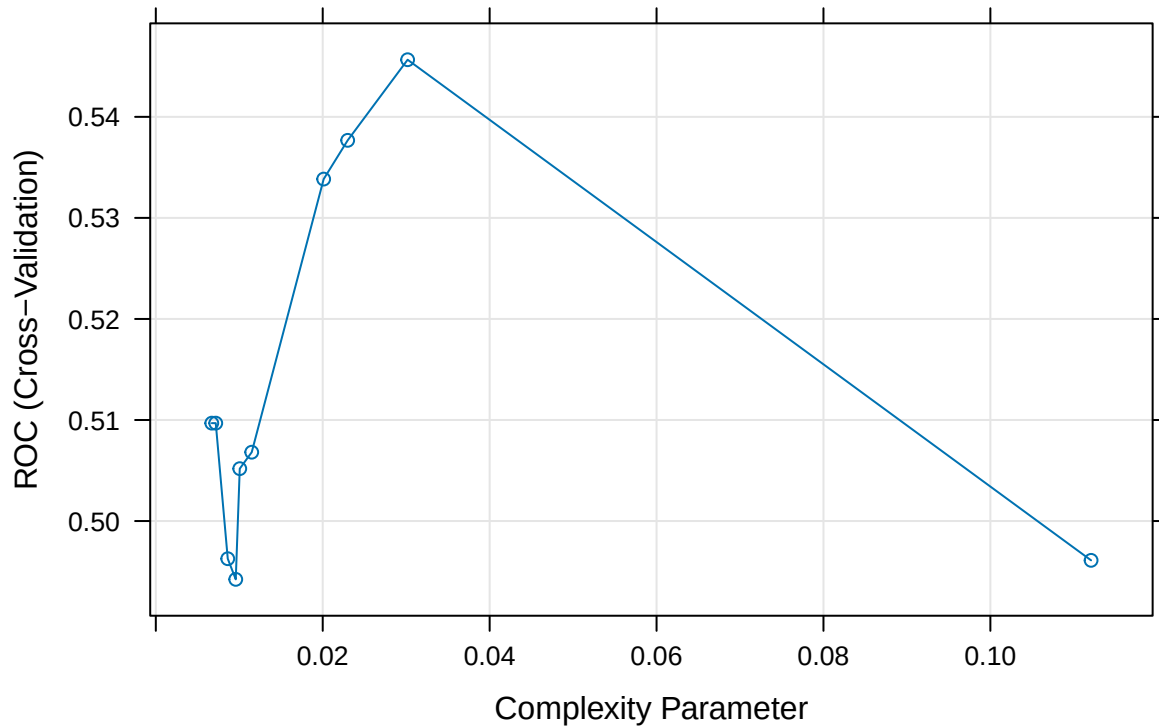
```
## Generalized Linear Model
##
## 701 samples
## 8 predictor
## 2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 630, 631, 631, 631, 631, 631, ...
## Resampling results:
##
## ROC      Sens      Spec
## 0.5151529 0.5296032 0.4511765
```

```
# Decision Tree with CV
tree_cv <- train(Churn ~ ., data=train,
                 method = "rpart",
                 trControl = ctrl,
                 tuneLength = 10,
                 metric = "ROC")
print(tree_cv)
```

```
## CART
##
## 701 samples
## 8 predictor
## 2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 630, 631, 630, 631, 631, 631, ...
## Resampling results across tuning parameters:
##
## cp      ROC      Sens      Spec
## 0.006704981 0.5096877 0.5016667 0.5054622
## 0.007183908 0.5096877 0.5016667 0.5054622
## 0.008620690 0.4962720 0.5335714 0.4594958
## 0.009578544 0.4942295 0.5363492 0.4565546
## 0.010057471 0.5051819 0.5553968 0.4651261
## 0.011494253 0.5068119 0.5582540 0.4854622
## 0.020114943 0.5338395 0.5664286 0.5087395
## 0.022988506 0.5376762 0.5835714 0.4973109
## 0.030172414 0.5456528 0.6036508 0.4889916
## 0.112068966 0.4961111 0.8293651 0.1628571
##
## ROC was used to select the optimal model using the largest value.
## The final value used for the model was cp = 0.03017241.
```

```
plot(tree_cv, main="Decision Tree Tuning Results")
```

Decision Tree Tuning Results

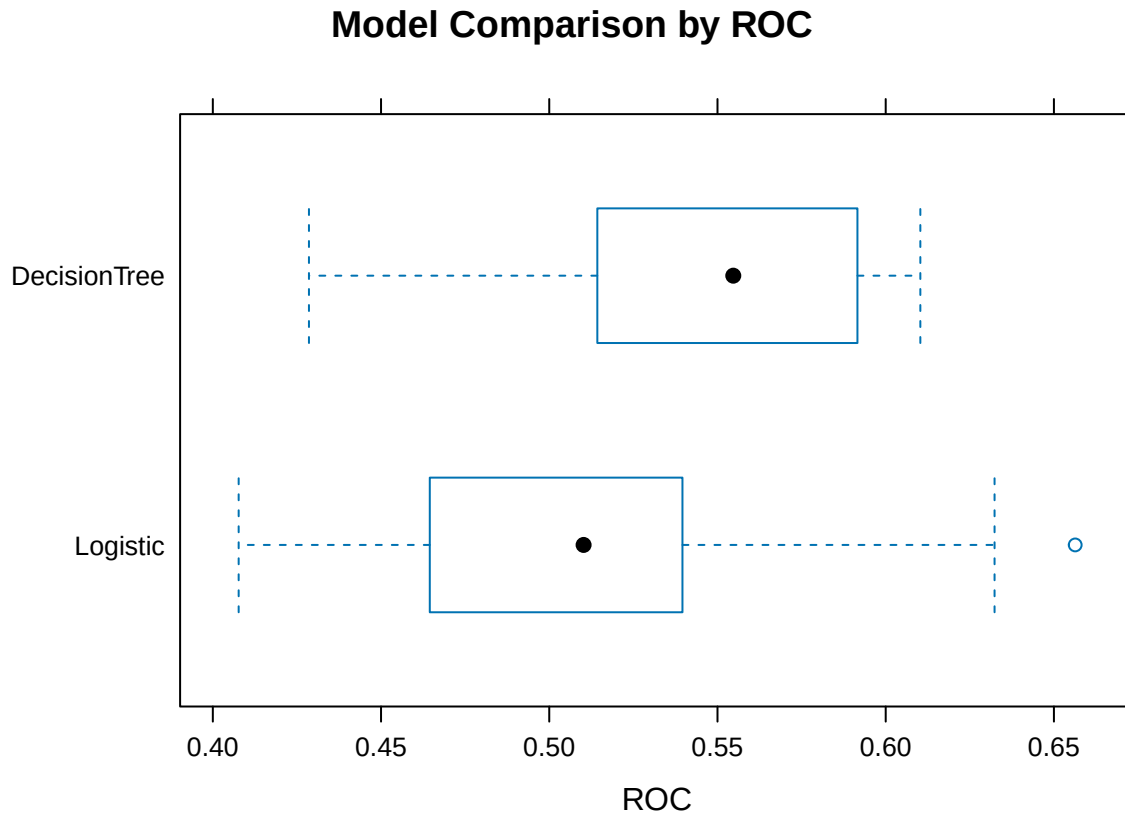


```
# Model comparison
models <- list(Logistic = logit_cv, DecisionTree = tree_cv)
resamples <- resamples(models)
summary(resamples)
```

```
##
## Call:
## summary.resamples(object = resamples)
##
## Models: Logistic, DecisionTree
## Number of resamples: 10
##
## ROC
##           Min.   1st Qu.   Median     Mean   3rd Qu.   Max. NA's
## Logistic  0.4076797 0.4656293 0.5102041 0.5151529 0.5377551 0.6563265    0
## DecisionTree 0.4285714 0.5156122 0.5546939 0.5456528 0.5841737 0.6102941    0
##
## Sens
##           Min.   1st Qu.   Median     Mean   3rd Qu.   Max. NA's
## Logistic  0.3428571 0.4928571 0.5416667 0.5296032 0.5674603 0.6571429    0
## DecisionTree 0.4571429 0.5138889 0.6285714 0.6036508 0.6809524 0.7428571    0
##
## Spec
```

##		Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
##	Logistic	0.2352941	0.3428571	0.4857143	0.4511765	0.5357143	0.6764706	0
##	DecisionTree	0.3142857	0.4071429	0.5142857	0.4889916	0.5808824	0.6571429	0

```
bwplot(resamples, metric = "ROC", main="Model Comparison by ROC")
```



Key findings (Model resampling):

Cross-validation Results

- Logistic Regression
 - ROC: 0.515
 - Sensitivity: 0.530
 - Specificity: 0.451
- Decision Tree (optimal cp = 0.030)
 - ROC: 0.546
 - Sensitivity: 0.604
 - Specificity: 0.489

Comparison The decision tree model shows a better overall performance with a higher ROC, sensitivity, and specificity. The box plot reinforces this by visualizing each models spread and mean value.

```
# Unsupervised Learning -----
```

```
# PCA Analysis
```

```
pca_data <- data %>% select(all_of(numeric_vars)) %>% scale()
```

```
pca_result <- PCA(pca_data, graph=FALSE)
```

```
summary(pca_result)
```

```
##
```

```
## Call:
```

```
## PCA(X = pca_data, graph = FALSE)
```

```
##
```

```
##
```

```
## Eigenvalues
```

```
##          Dim.1  Dim.2  Dim.3  Dim.4  Dim.5
## Variance      1.061   1.035   1.001   0.976   0.926
## % of var.      21.227  20.701  20.027  19.516  18.528
## Cumulative % of var. 21.227  41.928  61.955  81.472 100.000
```

```
##
```

```
## Individuals (the 10 first)
```

```
##          Dist  Dim.1  ctr  cos2  Dim.2  ctr  cos2  Dim.3
## 1          | 1.961 | 0.560 0.030 0.081 | -0.145 0.002 0.006 | 0.789
## 2          | 2.477 | 0.582 0.032 0.055 | -0.874 0.074 0.125 | 2.015
## 3          | 2.822 | -0.073 0.001 0.001 | 0.467 0.021 0.027 | 0.686
## 4          | 2.196 | 1.321 0.164 0.362 | 0.417 0.017 0.036 | -0.117
## 5          | 2.393 | -0.103 0.001 0.002 | -0.356 0.012 0.022 | 1.781
## 6          | 2.554 | -1.673 0.264 0.429 | 0.167 0.003 0.004 | -1.049
## 7          | 1.459 | 0.629 0.037 0.186 | -0.833 0.067 0.326 | 0.373
## 8          | 1.266 | -0.144 0.002 0.013 | 0.147 0.002 0.013 | 1.032
## 9          | 2.831 | -0.373 0.013 0.017 | -1.676 0.271 0.350 | 0.077
## 10         | 1.975 | -0.722 0.049 0.133 | 1.376 0.183 0.485 | -1.076
```

```
##
```

```
##          ctr  cos2
## 1          0.062 0.162 |
## 2          0.405 0.662 |
## 3          0.047 0.059 |
## 4          0.001 0.003 |
## 5          0.317 0.554 |
## 6          0.110 0.169 |
## 7          0.014 0.065 |
## 8          0.106 0.664 |
## 9          0.001 0.001 |
## 10         0.116 0.297 |
```

```
##
```

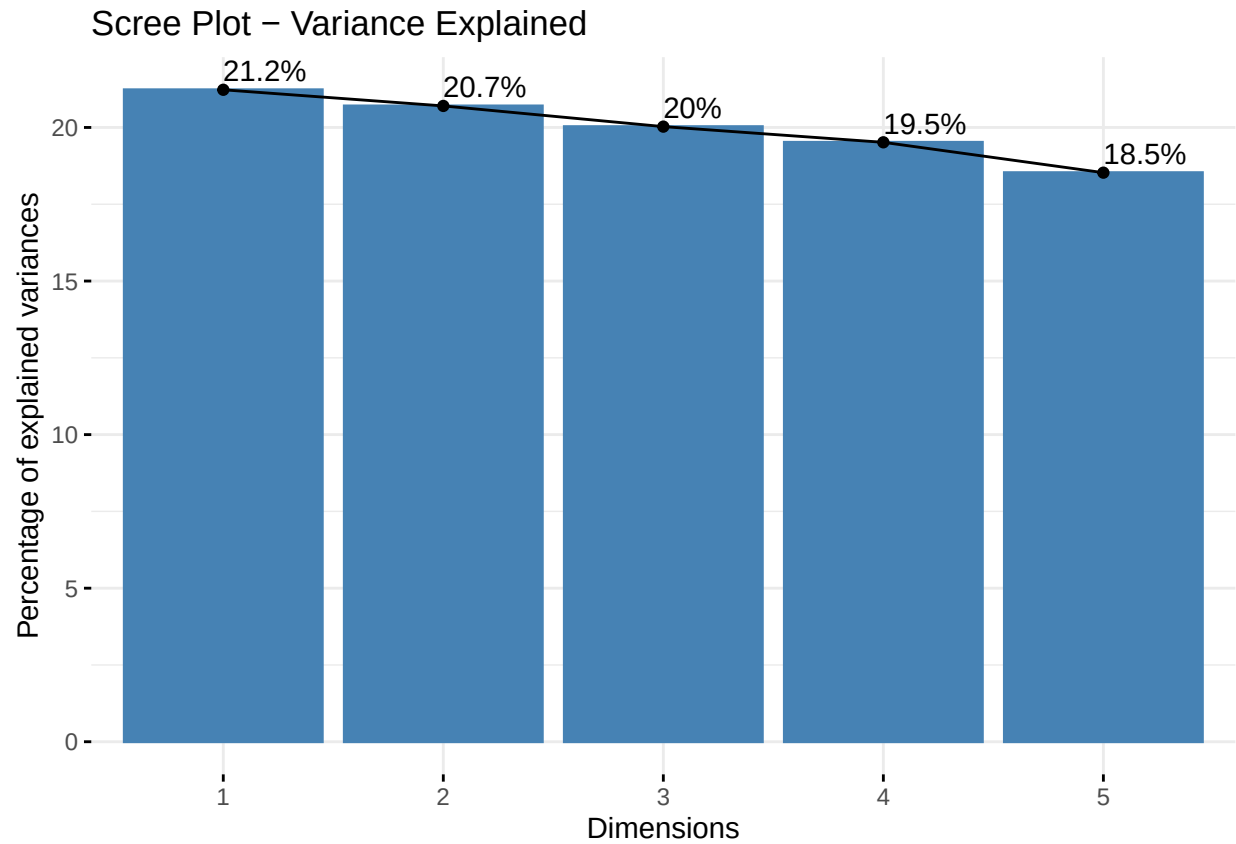
```
## Variables
```

```
##          Dim.1  ctr  cos2  Dim.2  ctr  cos2  Dim.3  ctr
## Age          | -0.125 1.477 0.016 | 0.371 13.286 0.138 | 0.831 68.998
## AnnualIncome | 0.198 3.677 0.039 | 0.613 36.313 0.376 | -0.515 26.528
## MonthlyCharges | 0.011 0.012 0.000 | 0.704 47.894 0.496 | 0.042 0.173
## TotalCharges | 0.693 45.264 0.480 | -0.157 2.382 0.025 | 0.166 2.750
## TenureMonths | 0.725 49.569 0.526 | 0.036 0.125 0.001 | 0.125 1.551
##          cos2
## Age          0.691 |
## AnnualIncome 0.266 |
## MonthlyCharges 0.002 |
```

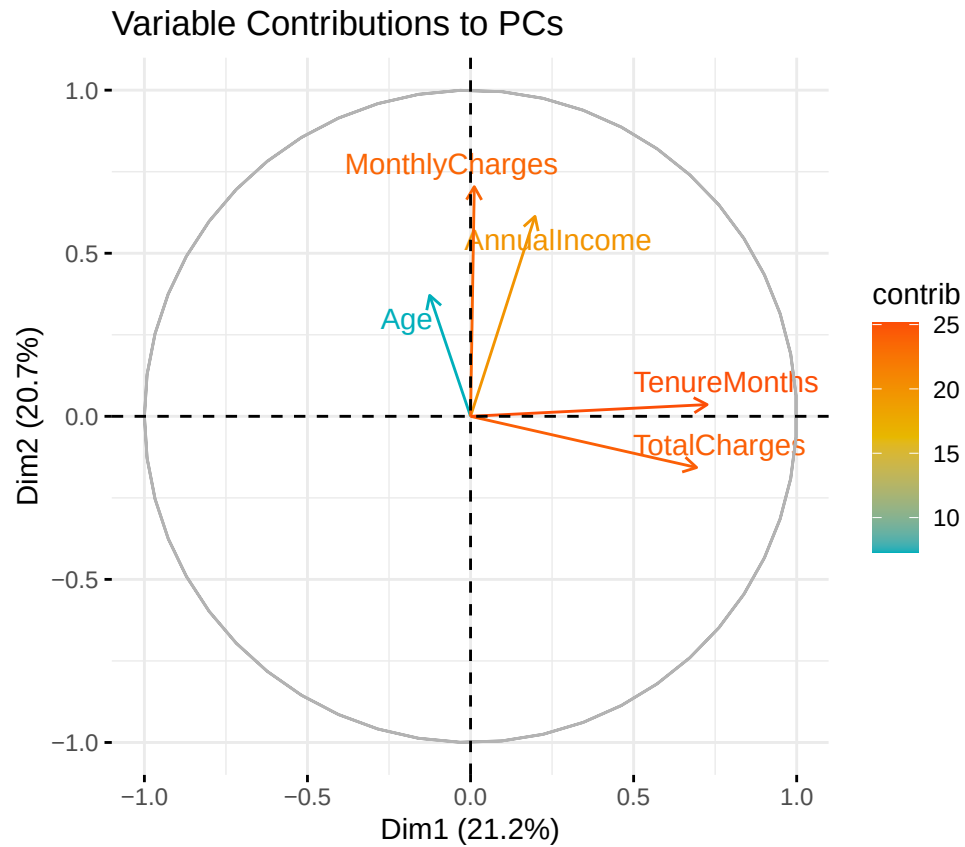
```
## TotalCharges    0.028 |  
## TenureMonths    0.016 |
```

```
# PCA Visualization
```

```
fviz_eig(pca_result, addlabels=TRUE, main="Scree Plot - Variance Explained")
```



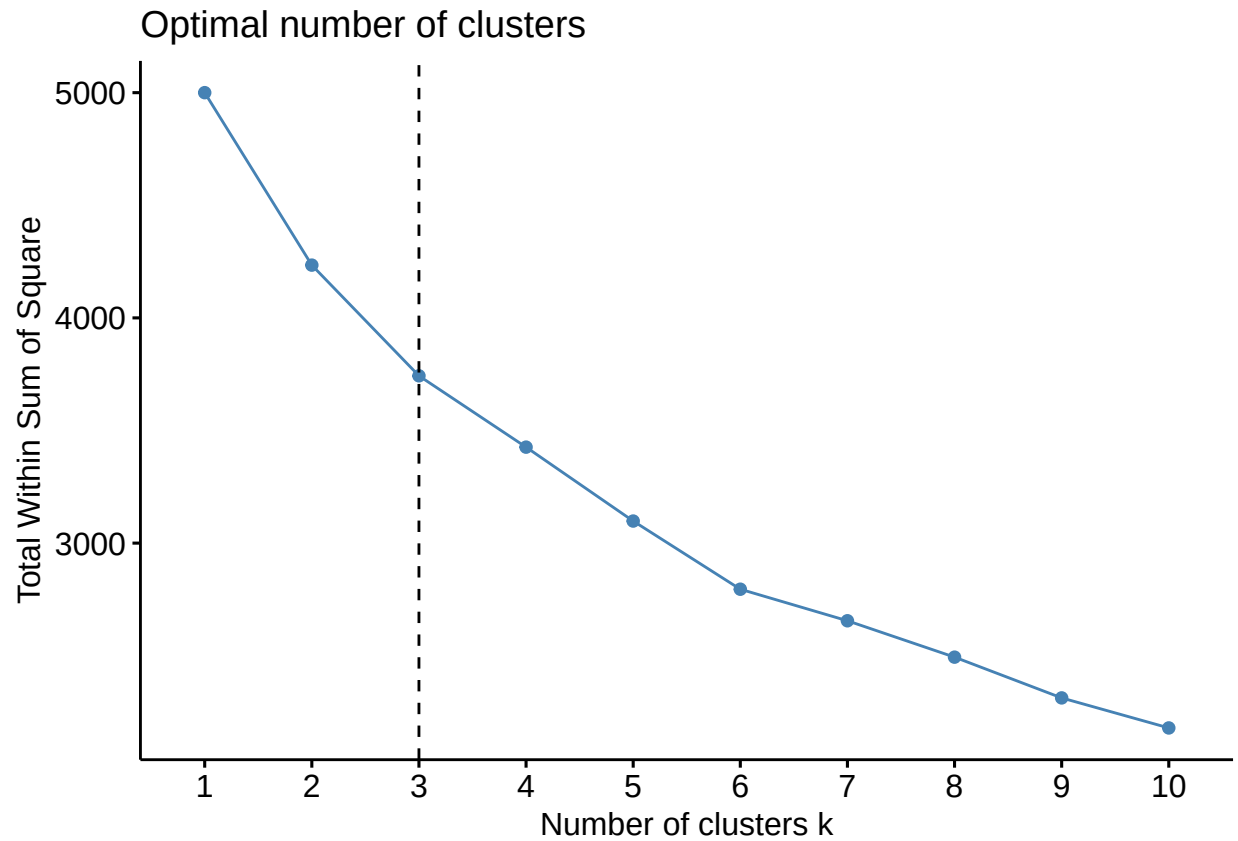
```
fviz_pca_var(pca_result, col.var="contrib",  
             gradient.cols=c("#00AFBB", "#E7B800", "#FC4E07"),  
             repel=TRUE, title="Variable Contributions to PCs")
```



```
# K-means Clustering with PCA components
pca_scores <- as.data.frame(pca_result$ind$coord)
colnames(pca_scores) <- paste0("PC", 1:ncol(pca_scores))

# Determine optimal clusters
fviz_nbclust(pca_scores, kmeans, method="wss") +
  geom_vline(xintercept = 3, linetype = 2) +

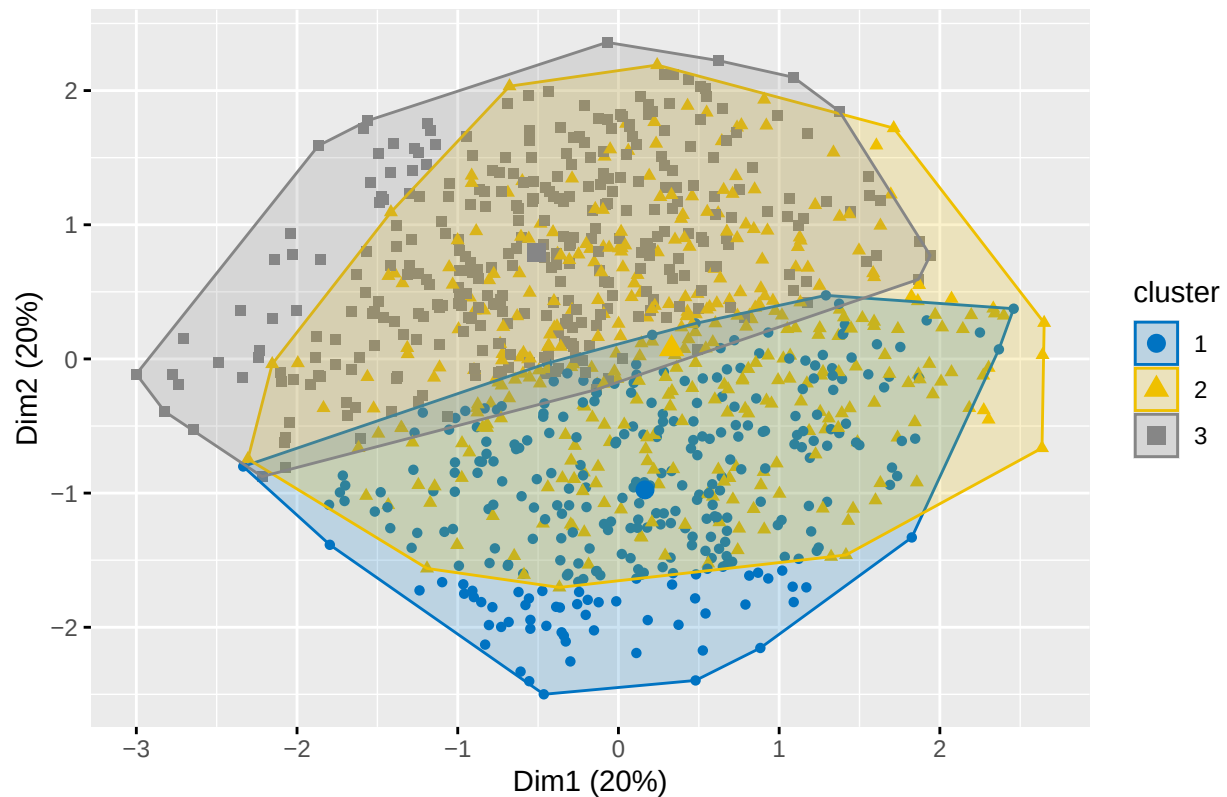
# Perform clustering
set.seed(123)
```



```
kmeans_result <- kmeans(pca_scores, centers=3, nstart=25)
data$Cluster <- as.factor(kmeans_result$cluster)

# Visualize clusters
fviz_cluster(kmeans_result, data=pca_scores,
  geom="point",
  ellipse.type="convex",
  palette="jco",
  main="Customer Segments from PCA")
```

Customer Segments from PCA



Analysis and Insight -----

Cluster analysis

```
cluster_summary <- data %>%
  group_by(Cluster) %>%
  summarise(
    Count = n(),
    ChurnRate = mean(Churn == "Yes"),
    AvgMonthlyCharge = mean(MonthlyCharges),
    AvgTenure = mean(TenureMonths),
    AvgAge = mean(Age),
    AvgIncome = mean(AnnualIncome)
  )

print(cluster_summary)
```

```
## # A tibble: 3 x 7
##   Cluster Count ChurnRate AvgMonthlyCharge AvgTenure AvgAge AvgIncome
##   <fct>   <int>   <dbl>         <dbl>    <dbl>  <dbl>   <dbl>
## 1 1       301     0.475          79.4     47.9   50.1   56004.
## 2 2       364     0.511          66.8     32.6   29.1   52069.
## 3 3       335     0.499          79.8     28.7   54.2   57013.
```

Key churn factors

```
varImp(logit_cv)
```

```
## glm variable importance
##
##           Overall
## AnnualIncome    100.000
## IsSeniorCitizen1 82.496
## NumOfProducts4   33.375
## TotalCharges     31.425
## MonthlyCharges   19.054
## Age              14.400
## TenureMonths      9.035
## NumOfProducts3    8.901
## NumOfProducts2    8.205
## HasTechSupportYes 0.000
```

```
varImp(tree_cv)
```

```
## rpart variable importance
##
##           Overall
## AnnualIncome    100.00
## MonthlyCharges   58.92
## TotalCharges     54.10
## IsSeniorCitizen1 44.95
## TenureMonths     36.99
## NumOfProducts2    0.00
## NumOfProducts3    0.00
## Age              0.00
## HasTechSupportYes 0.00
## NumOfProducts4    0.00
```

Key findings (PCA and K-means):

Principal Component Analysis

- The scree plot showed the first two components explained most variance
- Variable contribution plot indicated:
 - PC1 strongly influenced by tenure and charges
 - PC2 influenced by monthly charges, income, and age

K-means Clustering

- Cluster 1 (30.1% of customers):
 - Highest tenure (47.9 months)
 - Older customers (avg age 50.1)
 - Moderate churn rate (47.5%)
- Cluster 2 (36.4% of customers):
 - Younger customers (avg age 29.1)
 - Moderate tenure (32.6 months)
 - Highest churn rate (51.1%)
- Cluster 3 (33.5% of customers):

- Oldest customers (avg age 54.2)
- Shortest tenure (28.7 months)
- High monthly charges (\$79.80)
- Moderate churn rate (49.9%)

Summary and Recommendations

Critical Churn Factors

- Financial Factors:
 - Lower annual income customers are more likely to churn
 - Customers with higher monthly charges show higher churn tendencies
- Demographic Factors:
 - Senior citizens have significantly higher churn risk
 - Younger customers (Cluster 2) show the highest churn rates
- Service Factors:
 - Number of products didn't show strong predictive power
 - Tech support availability wasn't a significant factor

Strategic Recommendations

- Targeted Retention Programs:
 - Implement income-based pricing strategies for lower-income customers
 - Create senior citizen-specific service plans with added benefits
 - Develop loyalty programs for younger customers (Cluster 2)
- Service Optimization:
 - Review pricing structure for high monthly charge customers
 - Analyze why high-tenure customers (Cluster 1) still churn at 47.5% rate
- Proactive Engagement:
 - Implement early warning systems using the decision tree model
 - Focus outreach on customers matching high-risk cluster profiles
- Further Research:
 - Investigate why tech support isn't reducing churn as expected
 - Collect additional data on service quality and customer satisfaction
 - Explore more complex models to improve predictive accuracy

The analysis suggests that a segmented, data-driven approach focusing on financial and demographic factors will be most effective in reducing customer churn for this telecom company.